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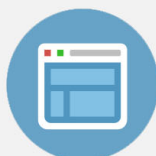
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An efficient microcanonical sampling procedure for molecular systems

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A general and efficient microcanonical sampling (EMS) procedure based on that of Severin *et al.* [Chem. Phys. Lett. **57**, 117 (1978)] is outlined. The relevant statistical weights are derived. The EMS method can be used for the generation of initial microcanonical states for trajectories or directly in the efficient calculation of microcanonical statistical averages and rate constants. The validity and usefulness of the EMS method is demonstrated in applications to 1D linear chains and 3D molecules such as H₂O, SiH₂, and Si₂H₆.

I. INTRODUCTION

It is frequently useful to be able to microcanonically sample the internal energy surface of a molecule (or molecular system) in an efficient and accurate manner. This has been of particular importance in the theoretical treatment and trajectory simulation of the intermolecular and intramolecular dynamics of unimolecular reactions.¹⁻²⁵ Initial steps in this direction were taken by Bunker¹⁻⁴ but the sampling algorithm used became very inefficient and was impractical for molecules larger than triatomics.² Essentially, Bunker's algorithm suffers from a low acceptance/rejection ratio in generating new states when applied to larger molecules.²⁶ An alternative procedure was outlined by Nordholm and co-workers^{26,27} and was found to be much more efficient than Bunker's algorithm.²⁶

Despite the success of this more accurate and potentially much more efficient procedure, there has been continued interest in the development of approximate methods of microcanonical sampling^{5,6,15-18} with varying degrees of success. Recently, a series of simulations was performed which further demonstrated the utility of the sampling method of Nordholm and co-workers²⁷ for the particular case of the dissociation of one-dimensional classical chain molecules with from two to ten atoms per molecule.¹⁹⁻²² The method has also been successfully applied in various forms to the sampling of diatomic and triatomic molecules²⁸⁻³¹ and to polyatomic molecules.^{32,33} Nyman *et al.*³⁰ have shown that the energy partitioning between potential and kinetic energy may be far from the harmonic approximation, especially for small molecules at high energies. This is relevant to approximate methods, such as orthant sampling^{3,6} which, while it has been shown to be exact for a system of independent harmonic oscillators,¹⁸ does not result in a truly microcanonical energy partitioning for more realistic systems.^{30,34} Using a momentum scaling parameter p_{scale} ,⁵ orthant sampling can be forced to yield the correct ratio of kinetic and potential energy if this ratio can be determined by independent means, e.g., trajectory calculation.⁵ However, this does not guarantee that other statistical averages and distributions will have

their true microcanonical values when generated using the orthant sampling method.

The point of this paper is to outline the general procedure by which both one-dimensional and three-dimensional molecules can be efficiently microcanonically sampled with respect to their internal energy. This efficient microcanonical sampling (EMS) procedure²⁷ is described in Sec. II and the form of the appropriate statistical weights is derived in Sec. III. Some demonstrations of the EMS procedure are described in Sec. IV and concluding remarks are made in Sec. V.

II. THE EFFICIENT MICROCANONICAL SAMPLING PROCEDURE

There is a problem associated with the growth of phase space with the number of atoms. For example, a 3-D N -atom system has a $6N$ -dimensional phase space. If it was necessary to sample phase space in some complete fashion, then the volume of numerical work would exceed the capability of available computers for quite modestly sized molecules. However, the success of Monte Carlo (MC)³⁵⁻³⁹ and molecular dynamics (MD)¹⁻⁶ methods in statistical mechanics and chemical reaction dynamics indicates that

(1) often it is not necessary to sample completely,

(2) MC and MD procedures search out the important parts of phase space in a very efficient manner.

The extent to which these favorable circumstances apply depends on the system and the properties studied. There is reason to be cautious with molecular models in this regard. However, the EMS method represents an attempt to exploit the great efficiency of the Monte Carlo sampling procedure with the aid of a Markovian stepping method in the coordinate part of phase space.

The sampling procedure to be described is based on that of Severin *et al.*^{26,27} It is described elsewhere in more detail for the particular case of one-dimensional classical chain molecules.^{19,20} Extension to three-dimensional molecules is trivial in principle. Thus, the algorithm will only be given a brief description. Let the molecular system to be sampled at an energy E , have momenta $\mathbf{p}$, spatial coordinates $\mathbf{q}$, and potential energy $V(\mathbf{q})$. For a molecule with N atoms, the vectors $\mathbf{p}$ and $\mathbf{q}$ are of dimension M where $M = N$ in one-dimensional space and $M = 3N$ in three-dimensional space. Let the molecular system have a Hamiltonian of the form

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$H = T(\mathbf{p}) + V(\mathbf{q})$, where T is the kinetic energy. The sampling procedure can then be divided into two parts, the selection of the spatial coordinates $\mathbf{q}$ and the selection of the momenta $\mathbf{p}$. For the selection of the spatial coordinates, it is convenient to specify the maximum ranges of these coordinates that are valid to describe the region of phase space that is of interest. In most studies¹⁻³³ the selection is limited to the reactant region. The spatial coordinates are then selected from a Markov chain generated by a Metropolis type procedure^{20,27,28,35} as follows:

(1) The molecule is set up in an arbitrary configuration, but preferably one that is fairly probable for the region of phase space to be sampled. For sampling from the reactant region of phase space, the molecule can be set up in an arbitrary orientation with equilibrium bond lengths. The statistical weight, $W_i = W(\mathbf{q}_i)$, of this initial configuration, $\mathbf{q}_i$, is then calculated. How this weight $W(\mathbf{q})$ is to be derived is described in Sec. III.

(2) A trial spatial configuration $\mathbf{q}_{\text{trial}}$ is generated from $\mathbf{q}_i$ by one of many possible procedures. For classical linear chains in one dimension,^{19,20} a bond was picked at random and its bondlength r was altered by a random perturbation $\pm \varepsilon$ (with random sign) where $0 \leq \varepsilon \leq \Delta r_{\text{max}}$ and Δr_{max} is the absolute magnitude of the maximum bond change allowed in each Markov chain step. The perturbed configuration thus generated is the trial spatial state. For systems in three dimensions, an acceptable procedure would be to pick an atom (or group of atoms) at random and move each atom by a random amount in a random direction. For examples of similar procedures see Thompson *et al.*⁷⁻¹²

(3) The statistical weight $W_{\text{trial}} = W(\mathbf{q}_{\text{trial}})$ of the trial spatial state $\mathbf{q}_{\text{trial}}$ is calculated.

(4) The trial spatial state $\mathbf{q}_{\text{trial}}$ is either accepted as a new state in the Markov chain or rejected according to the following criterion. If $W_{\text{trial}} \geq W_i$ then the trial state is accepted, else if $W_{\text{trial}} < W_i$ then a uniform random number R , where $0 \leq R \leq 1$, is compared with W_{trial}/W_i . If $W_{\text{trial}}/W_i > R$ then the trial state is accepted, otherwise it is rejected.

(5) If the trial state was accepted as the new state,

$$\mathbf{q}_{i+1} = \mathbf{q}_{\text{trial}} \quad (1)$$

and

$$W_{i+1} = W_{\text{trial}} \quad (2)$$

(6) If the trial state was rejected the old state is used as the new state,

$$\mathbf{q}_{i+1} = \mathbf{q}_i \quad (3)$$

and

$$W_{i+1} = W_i \quad (4)$$

The repetition of steps 2–6 then yields a Markov chain of spatial states $\{\mathbf{q}_1, \mathbf{q}_2, \dots\}$. However, they may be highly correlated so that, if uncorrelated states are desired, the states actually used for the next step, the selection of the momenta, should be a subset of these Markov chain states. One commonly used procedure³⁷⁻⁴⁰ is to allow an “incubation period” until the Markov chain states lose memory of their initial state. Then, periodically, a state is used for the selection

of the momenta, the period of selection being larger than the correlation length of the Markov chain. Graphical and empirical procedures for estimating the correlation lengths of Markov chains have been described elsewhere.^{19,39,40} Alternatively, all the states in the Markov chain can be used and correlation in the selected states can be accounted for when calculating the statistics.^{37,40-43}

Once a particular spatial configuration $\mathbf{q}$ has been selected from the Markov chain, it remains to specify the momenta $\mathbf{p}$ in order to have a fully described microcanonical state. The problem is now reduced to that of obtaining a microcanonical sample ensemble of momenta at the kinetic energy $E - V(\mathbf{q})$. The selection of the momenta is done according to a Gaussian distribution and these are scaled to the correct kinetic energy, $E - V(\mathbf{q})$.^{19,20,26,27} The vectors $\mathbf{p}$ and $\mathbf{q}$ are of dimension M where $M = N$ in one dimensional space and $M = 3N$ in three-dimensional space where N is the number of atoms in the system under consideration.

First, a vector $\hat{\mathbf{p}}$ of unscaled momenta is selected from a Gaussian distribution by a standard procedure.^{29,44} The unscaled velocities are

$$\hat{v}_i = \hat{p}_i / m_i, \quad i = 1, \dots, M, \quad (5)$$

where m_i is the mass of the atom associated with the i -th coordinate q_i . For an N -atom 1D chain, the unscaled center-of-mass velocity is

$$\hat{v}_{\text{c.m.}} = \sum_{i=1}^N m_i \hat{v}_i / \sum_{i=1}^N m_i, \quad (6)$$

with analogous expressions for the xyz components of the center-of-mass velocity for an N -atom 3D system. Thus, the unscaled internal kinetic energy

$$\hat{T}_{M,\text{int}} = \left(\sum_{i=1}^M \frac{1}{2} m_i \hat{v}_i^2 \right) - \frac{1}{2} m_{\text{tot}} \hat{v}_{\text{c.m.}}^2, \quad (7)$$

where m_{tot} is the total mass of the system. Noting that the correct internal kinetic energy is

$$T_{M,\text{int}} = E - V(\mathbf{q}), \quad (8)$$

a suitable scaling parameter α can be introduced as

$$\alpha = (T_{M,\text{int}} / \hat{T}_{M,\text{int}})^{1/2}. \quad (9)$$

The correctly scaled velocities and momenta are then

$$v_i = \alpha \hat{v}_i, \quad (10)$$

$$p_i = \alpha \hat{p}_i, \quad (11)$$

where $i = 1, \dots, M$. Thus, one now has a microcanonical state $(\mathbf{p}, \mathbf{q})$ at an internal energy E . The center-of-mass position and velocity are arbitrary but can be subtracted off to bring the molecular system into a Cartesian frame stationary with respect to and located at the center of mass.

III. THE STATISTICAL WEIGHTS

A. One-dimensional chains

It remains to specify the form of the statistical weight $W(\mathbf{q})$ to be associated with each spatial configuration in the spatial selection procedure described in Sec. II. From statistical mechanical considerations this weight is proportional to the internal density of states $\rho_{M,\text{int}}(E, \mathbf{q})$ at the configuration $\mathbf{q}$ with internal energy E .

It is illustrative to derive the required form of this weight for a one-dimensional chain of N atoms, for which $M = N$. The total Hamiltonian can be expressed as

$$H_{N,\text{tot}} = \sum_{i=1}^N \frac{p_i^2}{2m_i} + V(q_1, \dots, q_N). \quad (12)$$

It is convenient to transform to new spatial and momentum coordinates $(\mathbf{Q}, \mathbf{P})$ which separate out the center-of-mass contribution to the Hamiltonian in (12) as well as diagonalizing the remainder of this kinetic energy term. A general procedure for this would be to apply the F_2 type generator function,⁴⁵

$$F_2 = \sum_{i=1}^N f_i(\mathbf{q}, t) P_i, \quad (13)$$

where

$$p_j = \frac{\partial F_2}{\partial q_j}, \quad Q_j = \frac{\partial F_2}{\partial P_j}, \quad (14)$$

to a suitable $\mathbf{q} \rightarrow \mathbf{Q}$ coordinate transformation given by the functions $f_i(\mathbf{q}, t)$.

One source of suitable coordinate transformations is to employ the *mobile* method⁴⁶ described in detail by Hirschfelder.⁴⁷ A convenient spatial coordinate transformation of the *mobile* type is

$$Q_i = q_{i+1} - \sum_{j=1}^i m_j q_j / \sum_{j=1}^i m_j, \quad i = 1, \dots, N-1, \quad (15)$$

$$Q_N = \sum_{j=1}^N m_j q_j / \sum_{j=1}^N m_j. \quad (16)$$

This coordinate transformation can be written as

$$\mathbf{Q} = \mathbf{B}\mathbf{q}, \quad (17)$$

and from Eqs. (13) and (14) we get

$$\mathbf{p} = \mathbf{B}^T \mathbf{P}, \quad (18)$$

where

$$B_{ij} = \frac{\partial Q_i}{\partial q_j} = \frac{\partial p_j}{\partial P_i}. \quad (19)$$

The total Hamiltonian in Eq. (12) can now be written in the form

$$H_{N,\text{tot}} = H_{N,\text{int}} + T_{N,\text{ext}} \quad (20)$$

with the internal Hamiltonian as

$$H_{N,\text{int}} = \sum_{i=1}^{N-1} \frac{1}{2} G_{ii} P_i^2 + V(\mathbf{Q}) \quad (21)$$

and the external kinetic energy as

$$T_{N,\text{ext}} = \frac{1}{2} G_{NN} P_N^2, \quad (22)$$

where $\mathbf{G}$ is analogous to the familiar $\mathbf{G}$ matrix from spectroscopy as described by Wilson, Decius, and Cross,⁴⁸

$$G_{kl} = \sum_{i=1}^N \frac{1}{m_i} B_{ki} B_{li} \quad \text{for } k, l = 1, \dots, N. \quad (23)$$

The Jacobian for the transformation given by Eqs. (15)–(19) is

$$\begin{aligned} J(\mathbf{p}, \mathbf{q} | \mathbf{P}, \mathbf{Q}) &= J(\mathbf{p} | \mathbf{P}) J(\mathbf{q} | \mathbf{Q}) \\ &= J(\mathbf{p} | \mathbf{P}) [J(\mathbf{Q} | \mathbf{q})]^{-1} \\ &= |\mathbf{B}^T| [|\mathbf{B}|]^{-1} = 1, \end{aligned} \quad (24)$$

where we note that the inverse matrix $\mathbf{B}^{-1}$ must exist so that the determinant $|\mathbf{B}|$ is nonzero, and the total Jacobian is unity. It should be noted that our derivation of the statistical weights only requires that the total Jacobian is a finite constant.

By the separability of the total Hamiltonian into an internal and external part in Eq. (20), the total density of states $\rho_{N,\text{tot}}(E)$ can be separated into the internal density of states $\rho_{N,\text{int}}(E)$ and external density of states $\rho_{N,\text{ext}}(E)$ as follows:

$$\begin{aligned} \rho_{N,\text{tot}}(E) &= \frac{1}{h^N} \int dq_1 \cdots dq_N \\ &\quad \times \int dp_1 \cdots dp_N \delta(H_{N,\text{tot}} - E) \\ &= \int_0^E d\varepsilon \rho_{N,\text{int}}(\varepsilon) \rho_{N,\text{ext}}(E - \varepsilon), \end{aligned} \quad (25)$$

where

$$\begin{aligned} \rho_{N,\text{int}}(E) &= \frac{1}{h^{N-1}} \int dQ_1 \cdots dQ_{N-1} \\ &\quad \times \int dP_1 \cdots dP_{N-1} \delta(H_{N,\text{int}} - E) \end{aligned} \quad (26)$$

and

$$\rho_{N,\text{ext}}(E) = \frac{1}{h} \int dQ_N \int dP_N \delta(T_{N,\text{ext}} - E). \quad (27)$$

We can construct a transformation to scaled momenta $P_1, \dots, P_{N-1} \rightarrow \gamma_1, \dots, \gamma_{N-1}$, where

$$\gamma_i = [\frac{1}{2} G_{ii}]^{1/2} P_i, \quad i = 1, \dots, N-1, \quad (28)$$

so that

$$H_{N,\text{int}} = \sum_{i=1}^{N-1} \gamma_i^2 + V(\mathbf{Q}). \quad (29)$$

We can make a further transformation $\gamma_1, \dots, \gamma_{N-1} \rightarrow \gamma, \theta_1, \dots, \theta_{N-2}$ where γ is the magnitude of the vector $(\gamma_1, \dots, \gamma_{N-1})$ and the angles $\theta_1, \dots, \theta_{N-2}$ describe its orientation in $(N-1)$ space. Ignoring the angular dependence (which is of no consequence in this derivation and merely yields a constant factor after integration) the Jacobian for the total transformation is

$$J(P_1, \dots, P_{N-1} | \gamma, \theta_1, \dots, \theta_{N-2}) \propto \gamma^{N-2}. \quad (30)$$

Thus, from Eqs. (26) and (30),

$$\begin{aligned} \rho_{N,\text{int}}(E) &\propto \int dQ_1 \cdots dQ_{N-1} \\ &\quad \times \int d\gamma \gamma^{N-2} \delta(H_{N,\text{int}} - E) \end{aligned} \quad (31)$$

and

$$H_{N,\text{int}} = \gamma^2 + V(\mathbf{Q}). \quad (32)$$

By transforming $\gamma \rightarrow H_{N,\text{int}}$, and integrating over the delta function,

$$\rho_{N,\text{int}}(E) \propto \int dQ_1 \dots dQ_{N-1} [E - V(\mathbf{Q})]^{(N-3)/2}. \quad (33)$$

Thus, the required statistical weight for a one-dimensional chain of N atoms is proportional to the local density of states as

$$W(\mathbf{q}) \propto \rho_{N,\text{int}}(E, \mathbf{q}) \propto [E - V(\mathbf{q})]^{(N-3)/2}, \quad (34)$$

where the transformation back to $\mathbf{q}$ by Eqs. (15)–(19) merely introduces an arbitrary scaling factor which is independent of geometry.

B. Three-dimensional molecules

The above derivation can be analogously extended to a system of N atoms in three-dimensional space, for which $M = 3N$ for the selection algorithm described in Sec. II. The total Hamiltonian has the form,

$$H_{3N,\text{tot}} = \sum_{i=1}^{3N} \frac{p_i^2}{2m_i} + V(q_1, \dots, q_{3N}). \quad (35)$$

Following a route similar to that described by Eqs. (15)–(32), the internal density of states can be found to within an energy independent constant,

$$\rho_{3N,\text{int}}(E) \propto \int dQ_1 \dots dQ_{3N-3} [E - V(\mathbf{Q})]^{(3N-5)/2}, \quad (36)$$

so that the corresponding statistical weight for selecting configurations $\mathbf{q}$ for an N particle system in three dimensional space at an internal energy E is

$$W(\mathbf{q}) \propto \rho_{3N,\text{int}}(E, \mathbf{q}) \propto [E - V(\mathbf{q})]^{(3N-5)/2}. \quad (37)$$

Instead of working through the analogous derivation described above an alternative derivation will be described. Working directly from the total Hamiltonian in Eq. (35) we can use a similar series of steps to those leading from Eq. (26) to Eqs. (30)–(34) to yield the total density of states $\rho_{3N,\text{tot}}(E, \mathbf{q})$ at the configuration $\mathbf{q}$ with energy E ,^{26,27}

$$\rho_{3N,\text{tot}}(E, \mathbf{q}) = C_1 [E - V(\mathbf{q})]^{(3N-2)/2}, \quad (38)$$

where C_1 is a constant independent of E and $\mathbf{q}$. Because of the separability between the internal degrees of freedom and the external center-of-mass degrees of freedom,

$$\rho_{3N,\text{tot}}(E, \mathbf{q}) = \int_{V(\mathbf{q})}^E d\varepsilon \rho_{3N,\text{int}}(\varepsilon, \mathbf{q}) \rho_{3N,\text{ext}}(E - \varepsilon), \quad (39)$$

where the density of states of the external (center-of-mass) degrees of freedom is simply that for three independent translations,

$$\rho_{3N,\text{ext}}(E) = \frac{4\pi}{3} \Delta V \frac{\partial}{\partial E} [p_{\text{max}}^3(E)], \quad (40)$$

$$= 2\pi \Delta V (2M)^{3/2} E^{1/2}, \quad (41)$$

where

$$p_{\text{max}}(E) = (2ME)^{1/2}, \quad (42)$$

and ΔV is the volume available to the system and M is the total mass of the system.

The internal density of states $\rho_{3N,\text{int}}(E, \mathbf{q})$ at a configura-

tion $\mathbf{q}$ with internal energy E must be of the same form as Eq. (38), i.e.,

$$\rho_{3N,\text{int}}(E, \mathbf{q}) = C_2 [E - V(\mathbf{q})]^x, \quad (43)$$

where C_2 and x are constants independent of E and $\mathbf{q}$. Insertion of Eqs. (41) and (43) into Eq. (39) yields

$$\rho_{3N,\text{tot}}(E, \mathbf{q}) = C_3 \int_{V(\mathbf{q})}^E d\varepsilon [\varepsilon - V(\mathbf{q})]^x (E - \varepsilon)^{1/2}, \quad (44)$$

where

$$C_3 = C_2 2\pi \Delta V (2M)^{3/2}. \quad (45)$$

Making the substitution,

$$y = [\varepsilon - V(\mathbf{q})] / [E - V(\mathbf{q})], \quad (46)$$

in Eq. (44) yields

$$\begin{aligned} \rho_{3N,\text{tot}}(E, \mathbf{q}) &= C_3 \int_0^1 dy y^x (1 - y)^{1/2} [E - V(\mathbf{q})]^{x+3/2}. \end{aligned} \quad (47)$$

Comparison of Eq. (38) with Eq. (47) for all E leads to $x = (3N - 5)/2$ and thus, insertion into Eq. (43) leads to the desired statistical weight as given by Eq. (37).

IV. APPLICATIONS OF THE SAMPLING PROCEDURE

In order to apply the EMS procedure to the microcanonical sampling of a molecular system, the selection algorithm described in Sec. II must be used together with the appropriate statistical weights. For example, for a one-dimensional chain we use the weight in Eq. (34) and for a 3D molecule we use the weight in Eq. (37). The relationship between the 1D and 3D weights is clearer if we realize that the $N - 1$ internal momenta in the 1D case are replaced by the $(3N - 3)$ internal momenta in the 3D case. The weights (34) and (37) are consistent with those given by Severin *et al.*²⁷ for the sampling of a microcanonical ensemble at the total energy (including center-of-mass translation) of a 3D N -particle system,

$$W(\mathbf{q}) = [E_{\text{tot}} - V(\mathbf{q})]^{(3N-2)/2}, \quad (48)$$

where all $3N$ momenta are considered.

We will now describe some typical applications of the EMS procedure on some model 1D and 3D molecular systems. Statistical errors for the examples given are 95% confidence limits calculated from 5 to 10 independent batches of states by standard statistical techniques.⁴⁹ An alternative and useful nonparametric technique for estimating errors is the *bootstrap* method of Efron^{50,51} which has been employed with much success in assigning errors to trajectory derived rate constants¹⁹⁻²² and other averages.³⁴

As a first demonstration of the utility of the microcanonical sampling procedure we will consider an application to a model system, a one-dimensional chain of N atoms with identical masses and identical Morse bond potentials. The Hamiltonian is of the form of Eq. (12) with potential energy

$$V(q_1, \dots, q_N) = \sum_{i=1}^{N-1} \phi(q_{i+1} - q_i), \quad (49)$$

where

$$\phi(r) = D_e \{1 - \exp[-\beta_M(r - r_e)]\}^2. \quad (50)$$

We have chosen this extensively studied model (see references in Refs. 19–22) because of its simplicity and because it adequately demonstrates the main features of the sampling procedure. It is also a convenient choice as we have recently used the present sampling procedure for the same model system in a comparison of simulation with RRKM theory.^{19–22} The units are such that the dissociation energy $D_e = 1$, the particle mass $m_i = 1$, the equilibrium bond length $r_e = 1$ and the time unit is 1.47×10^{-13} s. The division between reactant and product space is given by the transition state which is placed at a somewhat arbitrary bond length r_c specified in a scaled form as $\beta_M(r_c - r_e)$. For more detail regarding this model see Ref. 20.

The selection procedure in Sec. II can be divided into a spatial coordinate selection procedure and a momentum coordinate selection procedure. The spatial part of the selection procedure requires that a Markov chain of spatial states be generated. In the case of the linear chain it is convenient to represent spatial configurations as a set of bond lengths. The efficiency with which a Markov chain explores the required configuration space is governed by how each Markov chain state is generated from the previous one.

In the case of our earlier linear chain studies^{19–22} a bond was picked at random and its length was changed by an amount $\pm \Delta r$ where

$$\Delta r = Rf(r_{\max} - r_{\min}), \quad (51)$$

where R is a uniform random number, $0 < R < 1$, f is a parameter which determines the largest fractional variation of a bond length allowed in a single Markov step, and $r_{\max}$ and $r_{\min}$ are the maximum and minimum allowed bondlengths, respectively. A small value of f leads to small Markov chain moves and a large value of f leads to large Markov chain moves. For molecular systems in 3D it is possible to perform the above spatial coordinate selection procedure in terms of internal coordinates, (e.g., bond lengths, bond angles) as has been demonstrated by Nyman *et al.*³⁰ but it is then necessary to modify the statistical weights by a factor resulting from the Jacobian transformation relating the Cartesian coordinates to the internal coordinates. An alternative and simpler procedure to implement for general systems is to pick an atom or group of atoms at random and perturb their Cartesian coordinates directly as commonly implemented in previous canonical and microcanonical sampling studies.^{7–12,35–38} The behaviour of a Markov chain is demonstrated in Fig. 1 for the simple case of a 1D uniform diatomic chain, for a large and small Cartesian step size, $\Delta_{\text{STEP}} = 0.5$ and 2.0 Å, respectively.

The number of Markov states generated before the Markov chain exhibits its typical periodicity is a measure of the incubation period while the periodicity thereafter is a measure of the correlation length.^{19,20,35} For a large enough number of successive Markov chain states N_{MCS} , the average probability of accepting a state P_{ACC} can be calculated from the statistical weight of the i -th state W_i and the statistical weight of the subsequent trial state W_{i+1}^{trial} ,

$$P_{\text{ACC}} = \frac{1}{N_{\text{MCS}}} \sum_{i=1}^{N_{\text{MCS}}} \min(W_{i+1}^{\text{trial}}/W_i, 1). \quad (52)$$

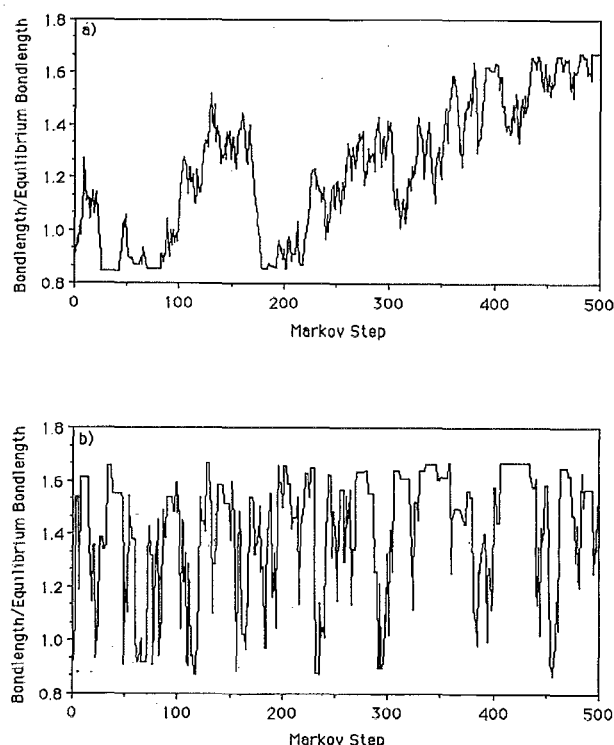


FIG. 1. Markov chain of bond lengths for a uniform diatomic chain generated by randomly perturbing the Cartesian coordinates of a random atom by up to $\pm \Delta_{\text{STEP}}$: (a) $\Delta_{\text{STEP}} = 0.5$ Å, (b) $\Delta_{\text{STEP}} = 2.0$ Å.

If N_{REJ} is the observed number of rejected states (trial states that are not accepted) then the observed rejection probability P_{REJ} can be calculated as

$$P_{\text{REJ}} = N_{\text{REJ}}/N_{\text{MCS}}. \quad (53)$$

If a sufficiently large number of states N_{MCS} is used (and depending on the quality of the pseudo random number generator employed in the algorithm)

$$P_{\text{ACC}} + P_{\text{REJ}} = 1 \quad (54)$$

should be satisfied to high accuracy and can be used as one of the tests that the algorithm is working correctly. This relation is verified in Table I for the simulation of linear chains.^{20–22}

TABLE I. Rejection and acceptance probabilities, P_{REJ} and P_{ACC} , respectively, as a function of Markov chain step size, given by f in Eq. (51), for 1D uniform chains with N atoms connected by Morse bonds (Refs. 19–22) at an internal energy $E = 1.5D_e$ with $\beta_M(r_c - r_e) = 10$. Five hundred Markov chain states were used.

N	f	P_{REJ}	P_{ACC}
2	0.1	0.062	0.938
	1.0	0.543	0.458
3	0.1	0.155	0.845
	1.0	0.698	0.302
4	0.1	0.295	0.706
	1.0	0.807	0.193
5	0.1	0.382	0.619
	1.0	0.867	0.133

TABLE II. Microcanonically averaged bondlengths $\langle r_i \rangle$ for 1D uniform chains with N atoms connected by harmonic or Morse bonds (Refs. 19–22). Internal energy $E = 1.5D_e$, $\beta_M(r_e - r_e) = 10$, Markov step size parameter $f = 1$, and 500 states were selected from the Markov chain. Errors quoted are 95% confidence limits.

Model	N	$\langle r_i \rangle / r_e$
Harmonic	2	0.984 ± 0.018
	3	0.989 ± 0.012 1.010 ± 0.012
	4	1.006 ± 0.010 0.994 ± 0.010 1.000 ± 0.010
	5	0.994 ± 0.008 1.000 ± 0.008 0.997 ± 0.010 1.000 ± 0.008
	5	0.994 ± 0.008 1.000 ± 0.008 0.997 ± 0.010 1.000 ± 0.008
Morse	2	2.125 ± 0.059
	3	1.587 ± 0.059 1.575 ± 0.059
	4	1.342 ± 0.051 1.380 ± 0.053 1.343 ± 0.050
	5	1.193 ± 0.037 1.203 ± 0.039 1.221 ± 0.041 1.229 ± 0.043
	5	1.193 ± 0.037 1.203 ± 0.039 1.221 ± 0.041 1.229 ± 0.043

From Fig. 1 and Table I we see that more rejections occur for large Markov chain moves where there are big differences in the configurations of successive Markov states. We have empirically found that the choice of Markov step size appears to be noncritical.¹⁹ However, much too small a step size will cause high acceptance rates of very correlated successive states and the correlation length will be very long. This is undesirable as the available configuration space will be explored relatively slowly and, if statistically independent states are desired, they will need to be selected from the Markov chain with a low frequency to minimize correlation with a commensurate increase in computer CPU time. A larger Markov step will result in a higher rejection rate but more configuration space is explored per Markov step and the correlation length is shorter. An unrealistically large Markov step will result in a high rejection rate, $P_{\text{REJ}} \approx 1.0$, and may result in an increase in the correlation length if the step size is increased without limit. It has been suggested^{35–38} that a 50% rejection rate is a good compromise for a reasonably short correlation length and a reasonably rapid exploration of configuration space.

In our past numerical simulations^{19–22} we have not rigorously ensured such optimization and have found Markov chains with rejection rates of 30–70% acceptable. It should be noted that, in applications where the selection procedure was used to provide initial states for trajectory simulation,^{20–22,28,29} it typically took a few percent of the total computation time while most of the time was used by the numerical integration of the equations of motion. In such applications it is not essential that the Markov stepping algorithm be optimized. However, it is much more important in applications where the microcanonical sampling procedure is used by itself to compute statistical averages, as in the Monte Carlo transition state theory (MCTST) calculations of Raff and Thompson.^{11,12,32,33}

As a demonstration of the selection procedure, the microcanonical average bond lengths of uniform classical chains of two to five atoms with $E = 1.5$ have been calculated. Table II gives the results for chains with harmonic bonds and Morse bonds. The harmonic chains have the expected average bond length of $r_e = 1.000$ within the 95% confidence limits quoted. The Morse bond lengths are of course at longer distances but approach the harmonic value as the average energy per bond $E/(N-1)$ decreases.

It must be stressed that in evaluating microcanonical averages which only depend on the spatial coordinates, such as bond lengths, we only need the spatial part of the selection procedure. However, to do molecular dynamics calculations we also need to specify the momenta via the method described in Sec. II. An illustrative example is the calculation of the microcanonical rate constant at short times for the decomposition of uniform classical chains. This should be equal to the RRKM rate constant if the dynamics have not had time to sufficiently perturb the initial microcanonical distribution.^{3,19–21} The short time results in Table III are for uniform chains at $E = 1.5$ and have been generated from 10^4 microcanonically selected initial states which were integrated for about 0.01–15% conversion into product. The average rate constant for short integration times ($t_{\text{max}} \ll 1/k$) is estimated as

$$k = \frac{-1}{t_{\text{max}}} \log_e [N(t_{\text{max}})/N(0)] \approx \frac{N(0) - N(t_{\text{max}})}{N(0)t_{\text{max}}}, \quad (55)$$

where $N(t_{\text{max}})/N(0)$ is the ratio of undissociated reactant states to initially selected states. Also shown in Table III are rate constants obtained from trajectories integrated for long times ($t_{\text{max}} \gg 1/k$). The rate constant is obtained in this case from a nonlinear least squares fit of the lifetime distribution obtained from 100–500 trajectories to the RRKM form^{19–22}

$$h(t) = ke^{-kt}. \quad (56)$$

It should be noted that other methods of obtaining rate constants from trajectory lifetimes are also available^{1–6,19–22} but will give similar results for the present application. The Morse RRKM rate constants shown in Table III take into account anharmonic effects as described elsewhere.^{19,20} The agreement between the RRKM and the short time simulation rate constants is generally within the estimated error in the simulation rate constants. Since few states dissociate at the shortest times considered the estimated errors are rather large despite the large number of trajectories integrated. At longer times, the estimated errors are lower but there appears to be some indication of non-RRKM behavior. Particularly evident in the smaller chains ($N \leq 5$) there is a trend for the simulation rate constant at the longer times to be larger than the RRKM predictions. This is likely to be due to slow IVR in the chain. In previous work on 1D chain systems,^{19–22} the time resolved rate constant $k(t)$ was generally found to start at the RRKM value at $t = 0$ and then rise

TABLE III. A comparison of microcanonical rate constants $k(E)$ calculated by short time and long time trajectory simulation with Morse RRKM predictions for 1D uniform chains with N atoms connected by Morse bonds (Refs. 19–22). Internal energy $E = 1.5D_e$, $\beta_M(r_c - r_e) = 10$. Markov step size parameter $f = 1$, and 10^4 or 100–500 microcanonically selected trajectories were run for the short and long integration times $t_{\max}$, respectively. Errors quoted are 95% confidence limits.

N	$t_{\max}/\text{ps}$	$k(E)/\text{ps}^{-1}$	
		Simulation	RRKM
2	0.0015	2.73 $\pm$ 0.84	2.17
	0.0074	2.17 $\pm$ 0.34	
	0.0147	2.34 $\pm$ 0.25	
	0.0735	2.40 $\pm$ 0.11	
	0.50	2.80 $\pm$ 0.07	
3	0.0015	0.885 $\pm$ 0.481	0.823
	0.0074	0.791 $\pm$ 0.203	
	0.0147	0.911 $\pm$ 0.154	
	0.0735	0.885 $\pm$ 0.067	
	7.35	1.18 $\pm$ 0.15	
4	0.0015	0.476 $\pm$ 0.353	0.450
	0.0074	0.491 $\pm$ 0.160	
	0.0147	0.533 $\pm$ 0.118	
	0.0735	0.495 $\pm$ 0.050	
5	0.0015	0.749 $\pm$ 0.442	0.248
	0.0074	0.395 $\pm$ 0.144	
	0.0147	0.403 $\pm$ 0.102	
	0.0735	0.360 $\pm$ 0.063	
	11.0	0.284 $\pm$ 0.049	
7	0.0015	0.068 $\pm$ 0.133	0.058
	0.0074	0.068 $\pm$ 0.060	
	0.0147	0.061 $\pm$ 0.040	
	0.0735	0.076 $\pm$ 0.020	
	73.5	0.046 $\pm$ 0.015	
10	0.0015	0.000	0.0036
	0.0074	0.000	
	0.0147	0.014 $\pm$ 0.019	
	0.0735	0.004 $\pm$ 0.005	
	147	0.0033 $\pm$ 0.0010	

above the RRKM value at short times before dropping below at long times. In principle, at short enough integration times $t_{\max}$ the RRKM prediction should be obtained. But at longer (but still short) times significant fast dissociations are occurring to raise the simulation rate constant above the RRKM value. Importance sampling of states near the transition state would enhance the efficiency of the above statistical-dynamical method of obtaining a microcanonical rate constant by enabling the use of much shorter integration times while still achieving a significant number of dissociations.

The use of microcanonically selected trajectories to calculate rate constants has the advantage that dynamical effects are properly included but the disadvantage that they can be rather time consuming, especially for larger molecular systems. However, in certain applications, statistical theories, such as RRKM⁵² or Doll's generalization⁵³ of Slater's Theory,⁵⁴ can be assumed valid. A suitable procedure to obtain the rate constant for reaction from a given potential surface is to employ a method such as Monte Carlo transition state theory (MCTST).^{9–12} This method avoids having

to run trajectories by instead performing a classical integral over phase space for the potential model under study using a Metropolis sampling technique.^{35–38}

Our sampling technique does not suffer from rejection to the degree that the method of Thompson *et al.*^{7–12} does and requires Markov sampling over only the configuration part of phase space,²⁷ so that it should be a viable and more efficient alternative. We have recently implemented and tested such a method in the calculation of the microcanonical rate constant for simple bond fissions in linear chains⁵⁵ and silicon containing compounds such as SiH₂ and Si₂H₆.^{32,33} Doll has shown⁵³ that the microcanonical rate constant $k(E)$ can be expressed as an average over the microcanonical ensemble,

$$k(E) = \frac{1}{2} \int d\Gamma \delta(H(\Gamma) - E) \delta(q_{RC} - q_C) |\dot{q}_{RC}| / \int d\Gamma \delta(H(\Gamma) - E), \quad (57)$$

where Γ is the complete set of position and momentum coordinates $\{\mathbf{q}, \mathbf{p}\}$, $H(\Gamma)$ is the Hamiltonian of the system excluding the center-of-mass motion, $q_{RC} = q_{RC}(\mathbf{q})$ is the reaction coordinate and q_C is the critical value required for reaction. Microcanonical rate constants have commonly been evaluated directly from the above equation by performing a Markov walk on a finite width energy shell, but this procedure is inefficient and approximate as the delta function must be broadened to keep the acceptance of new Markov states at a tolerable level.^{9,11} A much more easily evaluated form of the microcanonical rate constant can be derived by exploiting the separability between potential and kinetic energy,

$$H = T(\mathbf{p}) + V(\mathbf{q}), \quad (58)$$

and transforming Eq. (57) into^{32,33,55}

$$k(E) = \frac{1}{2} \int d\mathbf{q} W(\mathbf{q}) \delta(q_{RC} - q_C) \langle |\dot{q}_{RC}| \rangle_{MC,K} / \int d\mathbf{q} W(\mathbf{q}), \quad (59)$$

where $W(\mathbf{q})$ are the appropriate EMS statistical weights, Eqs. (34) or (37), for the system under study, and the microcanonical (MC) average absolute velocity through the critical surface,

$$\langle |\dot{q}_{RC}| \rangle_{MC,K} = \int d\mathbf{p} \delta(T - K) |\dot{q}_{RC}| / \int d\mathbf{p} \delta(T - K), \quad (60)$$

where

$$K = E - V(\mathbf{q}). \quad (61)$$

The EMS procedure can now be readily applied to the evaluation of $k(E)$ by Eq. (59) and $\langle |\dot{q}_{RC}| \rangle_{MC,K}$ is calculable from standard Monte Carlo sampling of the momenta.^{19,20,26} The calculation of microcanonical rate constants $k(E)$ by using EMS techniques in the transition state theory (TST) outlined above will be called EMS-TST and will be covered in more detail in forthcoming applications.^{32,33,55} A selection of typical results is shown in Table IV for the bond dissociation of a linear 5 atom chain at var-

TABLE IV. A comparison of EMS-TST (Refs. 32 and 55), trajectory simulation (Refs. 19–22), harmonic and Morse RRKM (Refs. 19 and 21) microcanonical rate constants for a 1D uniform chain with 5 atoms connected by Morse bonds. $\beta_M(r_c - r_e) = 6$. Errors quoted are 95% confidence limits.

E/D_e	EMS-TST			$k(E)/\text{ps}^{-1}$	Trajectory simulation	Harmonic RRKM	Morse RRKM
	(I)	(II)	(III)				
1.10	$(1.63 \pm 0.09) \times 10^{-2}$	$(1.17 \pm 0.06) \times 10^{-2}$	$(1.04 \pm 0.05) \times 10^{-2}$		$(9.93 \pm 5.10) \times 10^{-3}$	2.39×10^{-2}	1.12×10^{-2}
1.25	0.133 ± 0.002	$(9.55 \pm 0.18) \times 10^{-2}$	$(9.38 \pm 0.21) \times 10^{-2}$		$(6.05 \pm 1.97) \times 10^{-2}$	0.233	9.32×10^{-2}
1.50	0.459 ± 0.011	0.330 ± 0.008	0.316 ± 0.011		0.275 ± 0.054	1.05	0.326

ious energies. The EMS-TST results shown, numbered (I)–(III), employ three different definitions of the reaction coordinate $q_{RC}(\mathbf{q})$, which may be a function of some or all of the coordinates $\mathbf{q}$:

(I) One simple idea is to identify the reaction coordinate with the distance between the separating atoms (i.e., a bond length reaction coordinate⁵⁶) and the appropriate reduced mass is that corresponding to the two separating atoms.³²

(II) Identify the reaction coordinate with the bond length as in (I) but use a reduced mass corresponding to the two separating fragments.

(III) A very commonly used procedure⁵⁷ is to identify the reaction coordinate with the center-of-mass separation of the dissociating fragments and the appropriate reduced mass in this case is that corresponding to the separating fragments.³²

The statistical and trajectory results in Table IV are generated at the same bond length reaction coordinate separation, $\beta_M(r_c - r_e) = 6$, except for the EMS-TST(III) results which have been performed at a fragment–fragment center-of-mass separation which approximately corresponds to the same bond length separation. The EMS-TST(II) and EMS-TST(III) rate constants agree well with the Morse RRKM rate constants and reasonably well with the simulation rate constants. The EMS-TST(I) results predict rate constants, significantly larger than the EMS-TST(II), EMS-TST(III), and simulation rate constants, and perhaps represent a poor upper bound to the rate constant and a non-optimal choice of reaction coordinate. The harmonic RRKM predictions are larger than the other rate constants because of the neglect of anharmonicity. The major modifying effect of anharmonicity on the RRKM predictions is to increase the reactant density of states and thereby decrease the rate constant.^{19,21}

Nyman *et al.*³⁰ has previously applied a modification of the EMS procedure in the microcanonical sampling of H_2O , the major variation from the procedure outlined in the present paper was that the spatial selection procedure of Nyman *et al.*³⁰ did not use Markov sampling but relied on global Monte Carlo (GMC) acceptance/rejection methods.²⁹ Another difference was that the spatial coordinates were sampled in terms of bond lengths and bond angles. Thus, the appropriate statistical weight function for H_2O involved an extra factor of $R_a^2 R_b^2 \sin \alpha$ where R_a and R_b are bond lengths and α is the bond angle.³⁰ All coordinates were sampled independently so that a considerable amount of state rejection occurred. The advantage of the GMC method is that there is no incubation period or correlation length to consid-

er. Furthermore, if phase space is energetically divided into parts, a Markov chain procedure would need extra care to sample all of phase space while this is inherent to the GMC procedure. This, procedure could in principle be extended to higher polyatomics but would prove impractical because of further dramatic increases in state rejection.^{26,30} In such cases Markov sampling is the obvious method of choice.

We will compare the EMS method of Nyman *et al.*³⁰ which uses global Monte Carlo (GMC) sampling in its spatial selection procedure with the more generally applicable

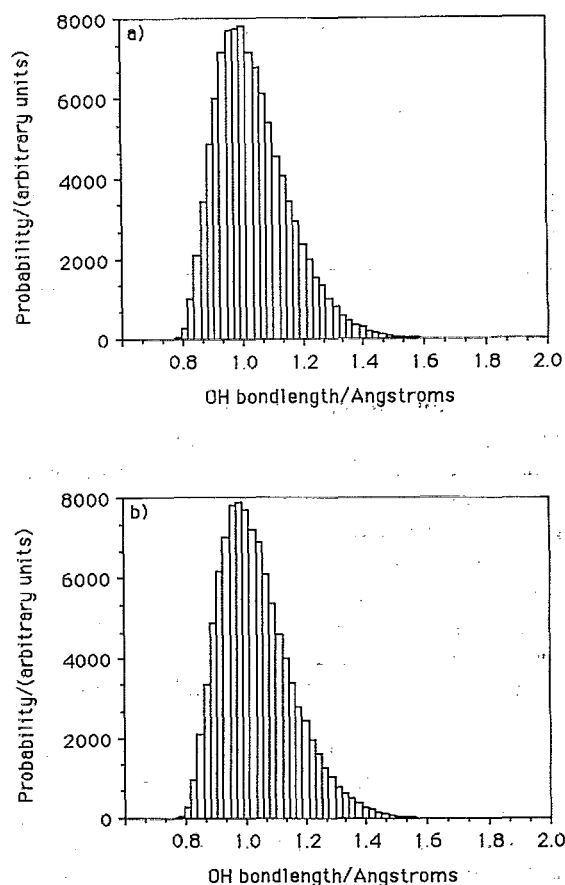


FIG. 2. Probability distribution of the OH bond length, $P(r_{\text{OH}})$, obtained by EMS on the PESIII model of H_2O of Nyman *et al.* (Ref. 30) using spatial selection procedures based on (a) global Monte Carlo sampling (Ref. 30), (b) Markov sampling (Refs. 19–22). The internal energy $E = 100$ kcal/mol and 10^5 states are used.

method of Markov (MK) sampling of spatial states. Results are shown in Figs. 2–5 and Table V for the PESIII model of H_2O constructed by Nyman *et al.*³⁰ The distributions and averages are identical within statistical error which further validates both procedures. The relative efficiency of the two methods can be gauged from the probability of state rejection P_{REJ} and the CPU time taken for the computation. For the results presented, it appears that there is little variation in the MK averages whether all states in the Markov chain are used or only every 10th or hundredth. If all states in the Markov chain are used, then the MK method is a factor of about 3.5–80 times more time efficient than the GMC method. Even if only every 100th state is used the MK method, while being a factor of 4 less efficient at low energies, is still much more efficient at high energies. The GMC method is very sensitive to the energy E , with far more rejection and computation time at higher energies, whereas the MK method takes about the same time at all energies studied. The rejection rate observed when using the GMC method depends on the form of the potential energy surface used, but the trends observed here should be fairly general. Thus, even for molecules as small as triatomics, it is apparent that Markov sampling of the spatial states has definite advantages. The computational advantage would be even more obvious for larger polyatomics.

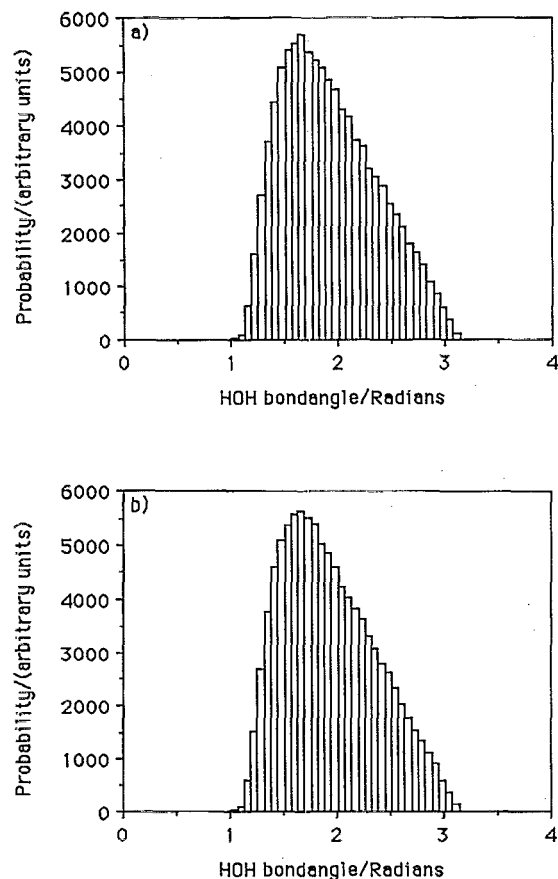


FIG. 3. As in Fig. 2 but for the probability distribution of the HOH bond angle, $P(\alpha_{\text{HOH}})$.

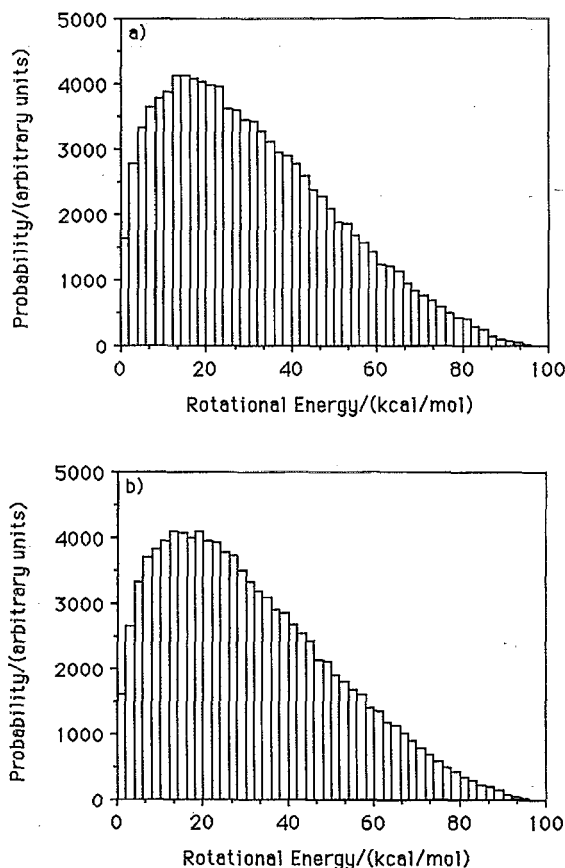


FIG. 4. As in Fig. 2 but for the probability distribution of the rotational energy E_{rot} , $P(E_{\text{rot}})$.

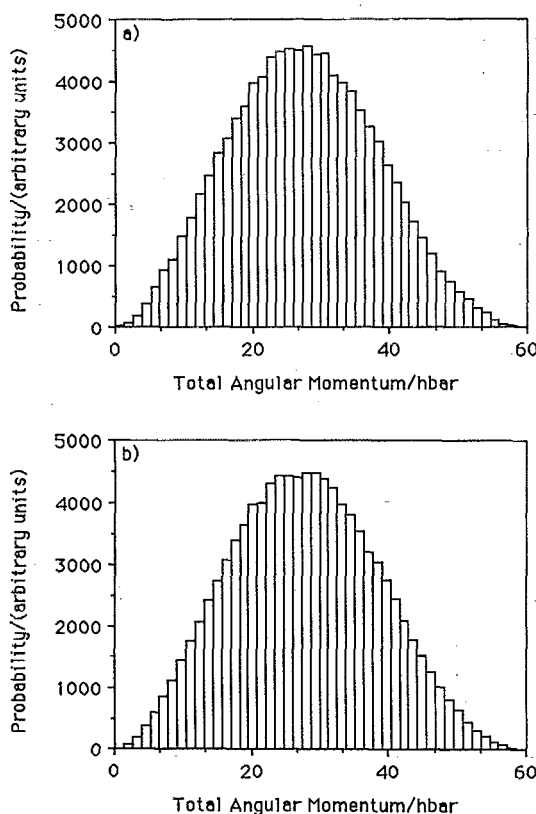


FIG. 5. As in Fig. 2 but for the probability distribution of the total angular momentum, L_{tot} , $P(L_{\text{tot}})$.

TABLE V. Comparison between selected averages generated by EMS using a spatial selection procedure based on global Monte Carlo (GMC) sampling and one based on Markov (MK) sampling for the PESIII model of H_2O (Ref. 30) as a function of internal energy E /(kcal/mol) and frequency of selection of states from the Markov chain, $1/N_F \times 10^5$ states are selected in all cases; the maximum Cartesian coordinate stepping parameter $\Delta_{\text{STEP}} = 0.05, 0.1, 0.3, 0.3 \text{ \AA}$ at $E = 1, 5, 100, 120$ kcal/mol, respectively. P_{REJ} = probability of state rejection; $\langle R_{\text{OH}} \rangle$ = average OH bond length in $\text{\AA}$; $\langle \alpha_{\text{HOH}} \rangle$ = average HOH bond angle in radians; $\langle E_{\text{rot}} \rangle$ = average rotational energy in kcal/mol, $\langle L_{\text{tot}} \rangle$ = average total angular momentum in $\hbar$ ($= \hbar/2\pi$); $\langle V \rangle/E$ = ratio of potential to internal energy; CPU = typical computation time on a Decstation-3100 workstation in seconds.

E	Method	N_F	P_{REJ}	$\langle R_{\text{OH}} \rangle$	$\langle \alpha_{\text{HOH}} \rangle$	$\langle E_{\text{rot}} \rangle$	$\langle L_{\text{tot}} \rangle$	$\langle V \rangle/E$	CPU
1.0	MK	100	0.806	0.9576	1.830	0.3334	2.697	0.3343	3253
	MK	50	0.806	0.9577	1.830	0.3331	2.696	0.3343	1583
	MK	1	0.807	0.9576	1.832	0.3316	2.688	0.3376	203
	GMC	...	0.991	0.9576	1.830	0.3320	2.678	0.3353	1280
5.0	MK	100	0.766	0.9593	1.857	1.652	6.011	0.3404	3331
	MK	50	0.767	0.9594	1.857	1.652	6.013	0.3404	1615
	MK	1	0.768	0.9594	1.860	1.652	6.015	0.3407	205
	GMC	...	0.982	0.9593	1.857	1.649	5.970	0.3403	740
100.0	MK	100	0.523	1.037	1.934	30.94	27.68	0.3826	3612
	MK	50	0.524	1.038	1.934	30.89	27.65	0.3839	1726
	MK	1	0.522	1.037	1.946	30.97	27.69	0.3820	208
	GMC	...	0.998	1.037	1.936	30.87	27.47	0.3833	4038
120.0	MK	100	0.479	1.083	1.865	34.09	29.75	0.4336	3623
	MK	50	0.479	1.083	1.866	34.12	29.74	0.4332	1729
	MK	1	0.482	1.086	1.862	33.54	29.58	0.4422	209
	GMC	...	0.9996	1.084	1.869	34.11	29.59	0.4320	17534

Further microcanonical sampling results are shown in Tables VI and VII for various statistical averages of models of SiH_2 and Si_2H_6 developed by Agrawal *et al.*^{58,59} The ratio of potential energy to internal energy $\langle V \rangle/E$ is a sensitive test of whether the sampling is truly microcanonical (though, of course, checks of other property averages and distributions should also be made). If we approximate the system under study by a model consisting of s independent harmonic oscillators and r independent free rotations then,

$$\langle V \rangle/E = s/(2s + r). \quad (62)$$

This model should be approximately valid for most potential models at low enough energy E . For a triatomic molecule we predict $\langle V \rangle/E = 0.333$ which compares well with the behavior shown for H_2O and SiH_2 in Tables V and VI at low energies. Similar agreement is shown for Si_2H_6 , where the prediction $\langle V \rangle/E = 0.462$ agrees with the low energy results in Table VII. The fact that $\langle V \rangle/E$ differs substantially from the predicted harmonic oscillator/rigid rotor equipartition at higher energies points to the importance of using an exact microcanonical sampling method rather than an approximate method.

TABLE VI. Average Si-H bond length $\langle r_{\text{SiH}} \rangle$ and ratio of potential energy to internal energy $\langle V \rangle/E$ generated by EMS for a model of SiH_2 (Ref. 58) as a function of internal energy. 5×10^6 states are used and reactant phase space is delimited by bond lengths $\leq 10.0 \text{ \AA}$. Errors quoted are 95% confidence limits.

E/eV	$\langle r_{\text{SiH}} \rangle/\text{\AA}$	$\langle V \rangle/E$
0.1	1.5017 ± 0.0001	0.345 ± 0.001
3.5	2.48 ± 0.08	0.569 ± 0.002
4.0	3.90 ± 0.17	0.756 ± 0.006
4.5	4.28 ± 0.18	0.738 ± 0.001

It is important to note that the present method is still viable when selecting internal states of a molecular system with further constraints other than the internal energy E . An obviously important case is the selection of (E, J) states where J is the angular momentum of the molecule. Such states are relevant to unimolecular reaction dynamics. Examples include the angular momentum resolved microscopic rate constant $k(E, J)$ ^{60,61} and the description of intermolecular energy transfer by angular momentum resolved transition probabilities $P(E', J' | E, J)$.^{29,62-64} Unfortunately, the same separation into spatial and momentum coordinates for the internal energy E , as in Eq. (21), does not generally apply for the angular momentum J which is usually coupled to both coordinates. The simplest solution is to specify a small range over J to be considered and apply an acceptance/rejection approach.²⁹⁻³¹ This is not absolutely necessary, if, as is common, $k(E, J)$ or $P(E', J' | E, J)$ are required for a whole range of E and J values. But for a given internal energy E , higher J values are far less common and accuracy will be far less for averages computed at high J unless one applies the above windowing technique. However, it is also possible to explicitly account for J conservation in the statistical weights. Work is in progress on an extension of the present method to sample microcanonically with respect to internal energy E and total angular momentum J .⁶⁵

V. CONCLUSIONS

An efficient algorithm for microcanonically sampling the internal energy surface of molecular systems has been presented which is much more efficient than other methods of comparable accuracy.^{26,27} This efficiency has been confirmed in several studies on 1D chain molecules^{19-22,55} and 3D models of SO_2 , H_2O , SiH_2 , and Si_2H_6 .²⁷⁻³³

The EMS method offers an efficient and, in principle, exact alternative to the many approximate and comparatively inefficient microcanonical sampling schemes still being

TABLE VII. Average Si-Si and Si-H bond lengths, $\langle r_{\text{SiSi}} \rangle$ and $\langle r_{\text{SiH}} \rangle$, and ratio of potential energy to internal energy $\langle V \rangle/E$ generated by EMS for a model of Si_2H_6 (Ref. 59) as a function of internal energy. 2.5×10^7 states are used and reactant phase space is delimited by bond lengths ≤ 10.0 Å. Errors quoted are 95% confidence limits.

E/eV	$\langle r_{\text{SiSi}} \rangle/\text{Å}$	$\langle r_{\text{SiH}} \rangle/\text{Å}$	$\langle V \rangle/E$
0.01	$2.381\,00 \pm 0.000\,07$	$1.478\,80 \pm 0.000\,04$	0.462 ± 0.002
0.1	$2.384\,1 \pm 0.000\,3$	$1.479\,7 \pm 0.000\,1$	0.466 ± 0.005
5.31	2.574 ± 0.008	1.550 ± 0.003	0.407 ± 0.001
6.31	2.718 ± 0.038	1.577 ± 0.003	0.419 ± 0.004
7.31	3.174 ± 0.046	1.642 ± 0.050	0.451 ± 0.008
8.31	3.763 ± 0.369	1.788 ± 0.086	0.503 ± 0.019
9.31	3.934 ± 0.294	2.005 ± 0.205	0.560 ± 0.035

employed in many simulations.⁵⁻¹⁸ A method with a somewhat similar motivation is that of Thompson *et al.*⁷⁻¹² However, they employ a prelimit form of the delta function over internal energy in Eq. (26) which is exactly resolved in our method. Thus, to ensure acceptance of states at a tolerable level, they have to allow some broadening in the energy of accepted states. Our method does not suffer from this source of rejection²⁷ and hence the efficiency of the Markov sampling in our method is likely to be much greater due to this factor alone, as well as the fact that only the spatial coordinates $\mathbf{q}$ are selected by Markov sampling in our method. The momenta $\mathbf{p}$ are selected without rejection using global Monte Carlo techniques.²⁶

Besides its use as a source of microcanonical initial states for trajectories^{19-22,28-31} and for the calculation of statistical averages (such as microcanonical rate constants^{32,33,55}), the EMS method could also be applied to the generation of anharmonic corrections to the molecular density of states. This is of much current interest as can be gauged from several recent and interesting studies.⁶⁶⁻⁶⁹

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